Carbon

Purpose

The carbon output files report the carbon budget of the soil, residue and plants. Two kinds of file are available:

- Flat files with one row per HRU or landscape unit per interval: carbon gains and losses, soil carbon transformations and plant carbon.
- Per-layer files with one row per HRU per interval and one column per soil layer for each variable: soil carbon totals, carbon pools, N and P in the pools, carbon and N fluxes, process drivers, pool dynamics and soil properties.

Every family has a row in print.prt with the usual daily, monthly, yearly and average annual flags. The rows are recognised only when print.prt uses object labels (use_obj_lbls = y); in the fixed-order layout there is no place for them. A row name that SWAT+ does not know stops the run with an error.

Most columns are filled by the dynamic carbon model, which runs when carbon = 2 in codes.bsn. Four per-layer families (hru_cb_cpool, hru_cb_flux, hru_cb_drv, hru_cb_dyn) are written only with carbon = 2. The other families are written for any setting, but with carbon = 0 many of their columns stay at zero. See the carbon model theory for the pools and transformations.

Files and switches

print.prt row	Files	Rows	Layers
hru_cb_gl	hru_carb_gl_{day,mon,yr,aa}	HRU	no
hru_cb_trf	hru_scf_{day,mon,yr,aa}	HRU	no
hru_cb_lyr	hru_cbn_lyr_{day,mon,yr,aa}	HRU	yes
hru_cb_cpool	hru_cpool_stat_{day,mon,yr,aa}	HRU	yes, carbon = 2 only
hru_cb_npool	hru_n_p_pool_stat_{day,mon,yr,aa}	HRU	yes
hru_cb_plt	hru_plc_stat_{day,mon,yr,aa}	HRU	no
hru_cb_flux	hru_cflux_stat_{day,mon,yr,aa}	HRU	yes, carbon = 2 only
hru_cb_drv	hru_carb_drv_{day,mon,yr,aa}	HRU	yes, carbon = 2 only
hru_cb_dyn	hru_carb_dyn_{day,mon,yr,aa}	HRU	yes, carbon = 2 only
hru_cb_snap	hru_soil_snap_{day,mon,yr,tot} and hru_soil_snap_tot	HRU	yes
lsu_cb_gl	lsu_carb_gl_{day,mon,yr,aa}	landscape unit	no
lsu_cb_trf	lsu_scf_{day,mon,yr,aa}	landscape unit	no
lsu_cb_plt	lsu_plc_stat_{day,mon,yr,aa}	landscape unit	no

Each file is written as .txt, plus .csv when csvout = y. The hru_cb_snap row writes _tot in place of _aa; see Soil snapshot. The landscape unit files are written only when the model defines landscape units in ls_unit.def.

Example print.prt rows (labelled layout):

```
objects                daily      monthly       yearly        avann
hru_cb_gl                  n            y            y            y
hru_cb_lyr                 n            n            y            y
hru_cb_snap                n            n            n            y
```

Value types and aggregation

The flat gain and loss files (`hru_carb_gl`, `hru_scf`, `lsu_carb_gl`, `lsu_scf`) report fluxes. Monthly and yearly files are sums over the interval, and the average annual file divides the run total by the number of printed years.

All other files report a state at the moment the row is written: the monthly file gives the value on the last day of the month, the yearly file the value on the last day of the year, and the average annual file the value at the end of the run. They are not averages. This also applies to the per-layer flux file `hru_cflux_stat`: its daily, monthly and yearly rows give the flux of the last day of the interval in kg/ha/day, and only the average annual file gives the run total divided by the number of printed years, in kg/ha/yr.

Flat HRU files

Carbon gains and losses

`hru_carb_gl_*` (row `hru_cb_gl`). One row per HRU. After the seven leading columns (`jday`, `mon`, `day`, `yr`, `unit`, `gis_id`, `name`) come 21 columns in kg C/ha, in three groups.

Soil:

Column	Description
`sed_c`	carbon removed with eroded sediment
`surq_c`	dissolved carbon in surface runoff
`latq_c`	dissolved organic carbon in lateral flow, all layers
`perc_c`	dissolved carbon in percolation below the profile
`res_decay`	carbon added to the soil by residue decay
`man_app_c`	carbon added with applied manure
`man_graze_c`	carbon added with manure from grazing animals
`rsp_c`	CO_2-C from soil respiration, whole profile
`soil_emit_c`	CO_2-C from burning soil carbon

Residue:

Column	Description
`plant_surf_c`	carbon added to surface residue by leaf drop and kill
`plant_root_c`	carbon added to soil residue by root death
`rsd_surfdecay_c`	carbon lost from surface residue by decay
`rsd_rootdecay_c`	carbon lost from root and incorporated residue by decay
`harv_stov_c`	carbon removed by residue harvest
`rsd_emit_c`	CO_2-C from burning surface residue

Plant:

Column	Description
`npp_c`	plant carbon growth (net primary production)
`harv_abgr_c`	carbon removed by grain and biomass harvest
`harv_root_c`	carbon removed by tuber (root) harvest
`drop_c`	carbon moved to residue by leaf drop and kill
`grazeat_c`	carbon eaten by grazing animals
`plant_emit_c`	CO_2-C from burning plants

Soil carbon transformations

`hru_scf_*` (row `hru_cb_trf`). One row per HRU with 13 columns in kg C/ha: `meta_micr`, `str_micr`, `str_hs`, `co2_meta`, `co2_str`, `micr_hs`, `micr_hp`, `hs_micr`, `hs_hp`, `hp_micr`, `co2_micr`, `co2_hs`, `co2_hp`. Each name gives the source and destination pool: `meta` metabolic litter, `str` structural litter, `micr` microbial biomass, `hs` slow humus, `hp` passive humus, and `co2` carbon lost as CO_2.

In 62.0.1 these columns are reset to zero each day and never filled, so every value in `hru_scf_*` and `lsu_scf_*` is zero. For the same flows by layer, use `hru_cflux_stat` (`cfmets1` for `meta_micr`, `co2fs1` for `co2_micr`, and so on).

Plant carbon

`hru_plc_stat_*` (row `hru_cb_plt`). One row per HRU with seven columns in kg C/ha:

Column	Description
`total_c`	carbon in all plant parts
`ab_gr_c`	carbon in above-ground biomass
`leaf_c`	carbon in leaves
`stem_c`	carbon in stems
`seed_c`	carbon in seeds
`root_c`	carbon in roots
`surf_rsd_c`	carbon in residue

Per-layer HRU files

Layout

Each row starts with the seven leading columns, then a depth block `depth_lyr1` to `depth_lyrN` giving the depth to the bottom of each soil layer in mm, then one block of N columns for each variable, named `<variable>_lyr1` to `<variable>_lyrN`. There is no units line: line 1 is the project name and version, line 2 the column header, and the data start on line 3.

N is the same for every HRU. By default it is the largest number of soil layers among all HRUs, so the deepest profile is written in full. The optional file `carbon_layers.prt` sets N directly. In an HRU with fewer layers than N, the extra columns contain `-99`.

Soil carbon by layer

`hru_cbn_lyr_*` (row `hru_cb_lyr`). Line 1 ends with the note `total soil carbon (Mg/ha) by layer depth in mm`. The value columns are in Mg C/ha:

Columns	Description
`depth_lyr1` … `depth_lyrN`	depth to the bottom of each layer (mm)
`tot_300_sum`	total soil organic carbon in the top 300 mm
`tot_lyr1` … `tot_lyrN`	total soil organic carbon in each layer
`seq_300_sum`	sequestered (humus and microbial) carbon in the top 300 mm
`seq_lyr1` … `seq_lyrN`	sequestered carbon in each layer

For the 300 mm sums, the layer that crosses 300 mm is counted in proportion to its thickness above 300 mm. With `carbon = 0`, both totals are the sum of the active humus, stable humus and microbial carbon.

Carbon pools

`hru_cpool_stat_*` (row `hru_cb_cpool`, `carbon = 2` only). Ten variables per layer:

Variable	Units	Description
`residue_c`	kg C/ha	residue carbon in the layer
`structural_c`	kg C/ha	structural litter
`metabolic_c`	kg C/ha	metabolic litter
`hs_c`	kg C/ha	slow humus
`hp_c`	kg C/ha	passive humus
`microbial_c`	kg C/ha	microbial biomass
`lignin_c`	kg C/ha	lignin part of structural litter
`nonlignin_c`	kg C/ha	non-lignin part of structural litter

Variable	Units	Description
root_mass	kg/ha	root mass in the layer
soil_water	mm	water in the layer

N and P in the pools

hru_n_p_pool_stat_* (row hru_cb_npool). Eighteen variables per layer, in kg N/ha and kg P/ha: tot_pool_n, residue_n, structural_n, metabolic_n, hs_n, hp_n, microbial_n, lignin_n, nonlignin_n, then the same nine with _p. tot_pool_n and tot_pool_p add residue, structural and metabolic litter, slow and passive humus, microbial biomass, the dissolved organic pool and manure.

Carbon and N fluxes

hru_cflux_stat_* (row hru_cb_flux, carbon = 2 only). Thirty-seven variables per layer. In the names, s1 is microbial biomass, s2 slow humus, s3 passive humus, met metabolic litter and str structural litter, so cfs1s2 is the carbon flow from microbial biomass to slow humus.

Variables	Units	Description
cfmets1, cfstrs1, cfstrs2	kg C/ha/day	carbon moved from litter to microbial biomass or slow humus
efmets1, efstrs1, efstrs2	kg N/ha/day	nitrogen moved with those flows
immmets1, immstrs1, immstrs2	kg N/ha/day	N immobilised by the litter flows
mnrmets1, mnrstrs1, mnrstrs2	kg N/ha/day	N mineralised by the litter flows
co2fmet, co2fstr	kg C/ha/day	CO_2-C from metabolic and structural litter
cfs1s2, cfs1s3, cfs2s1, cfs2s3, cfs3s1	kg C/ha/day	carbon moved between the humus and microbial pools
efs1s2, efs1s3, efs2s1, efs2s3, efs3s1	kg N/ha/day	nitrogen moved with those flows
imms1s2, imms1s3, imms2s1, imms2s3, imms3s1	kg N/ha/day	N immobilised by those flows
mnrs1s2, mnrs1s3, mnrs2s1, mnrs2s3, mnrs3s1	kg N/ha/day	N mineralised by those flows
co2fs1, co2fs2, co2fs3	kg C/ha/day	CO_2-C from microbial biomass, slow humus and passive humus

The average annual file gives yearly totals (kg/ha/yr); see Value types and aggregation.

Process drivers

hru_carb_drv_* (row hru_cb_drv, carbon = 2 only). Fourteen variables per layer, all for the day the row is written:

Variable	Units	Description
sut	fraction	soil water control on decomposition
tillagef	-	tillage factor of the layer
cons_bmix	-	biological mixing coefficient of the layer
tillagef_biomix	-	part of the tillage factor from biological mixing
tillagef_tillmix	-	part of the tillage factor from tillage operations
till_eff	fraction	tillage effect on decomposition
cdg	fraction	soil temperature control on decomposition
ox	fraction	oxygen control on decomposition
cs	fraction	combined control on decomposition
no3	kg N/ha	nitrate available to the transformations
nh4	kg N/ha	ammonium available to the transformations
co2_resp	kg C/ha	CO_2-C respired in the layer
soil_temp	°C	soil temperature
emix	fraction	fraction of the layer mixed by tillage or biological mixing

Pool dynamics

hru_carb_dyn_* (row hru_cb_dyn, carbon = 2 only). Twenty-one variables per layer:

Variables	Units	Description
asp, abp	fraction	share of decomposed slow humus and of decomposed microbial biomass sent to passive humus
abco2, a1co2, asco2, apco2	fraction	share of decomposed microbial biomass, metabolic and passive pools, slow humus and passive humus lost as CO_2
ncbm, nchp, nchs	ratio	N:C ratio of microbial biomass, passive humus and slow humus
bmctp, bmntp	kg/ha/day	potential C and N transformation of microbial biomass
hsctp, hsntp	kg/ha/day	potential C and N transformation of slow humus
hpctp, hpntp	kg/ha/day	potential C and N transformation of passive humus
lmctp, lmntp	kg/ha/day	potential C and N transformation of metabolic litter
lsctp, lslctp, lslnctp, lsntp	kg/ha/day	potential transformation of structural litter: C, lignin C, non-lignin C and N

Soil snapshot

hru_soil_snap_* (row hru_cb_snap). Thirteen soil properties per layer:

Variable	Units	Description
bd	Mg/m^3	bulk density
awc	mm/mm	available water capacity
soil_k	mm/h	saturated hydraulic conductivity
tot_c	kg C/ha	total soil organic carbon
clay, silt, sand, rock	%	particle size fractions and rock fragments
alb	fraction	albedo
usle_k	-	USLE soil erodibility
ec	dS/m	electrical conductivity
caco3	%	calcium carbonate content
ph	-	soil pH

The daily, monthly and yearly flags write hru_soil_snap_day, _mon and _yr. The average annual flag writes hru_soil_snap_tot instead, which has one row per HRU at the start of the run (marked jday = 0) and one at the end. Both rows give tot_c in kg C/ha, so the change in soil carbon in layer K over the run is tot_c_lyrK on the end row minus tot_c_lyrK on the start row. The start rows are written only when carbon = 2.

Repeated rows

In 62.0.1 the files of the hru_cb_lyr, hru_cb_cpool, hru_cb_npool, hru_cb_plt, hru_cb_flux, hru_cb_drv and hru_cb_dyn rows, and the daily, monthly and yearly files of hru_cb_snap, repeat their block of HRU rows once per HRU. In a model with 12 HRUs, every interval has 144 rows: the 12 HRU rows written 12 times with the same values. Keep the first occurrence of each unit within an interval when reading these files. hru_soil_snap_tot, hru_carb_gl, hru_scf and the landscape unit files are not affected.

Landscape unit files

The landscape unit files are area-weighted averages of the HRU values, each HRU weighted by its fraction of the landscape unit. Per-layer files have no landscape unit version because the HRUs in a unit can have different soil profiles.

File	Columns
lsu_carb_gl_*	the 21 columns of hru_carb_gl
lsu_scf_*	the 13 columns of hru_scf (all zero in 62.0.1)
lsu_plc_stat_*	one column, lsu_plt_c: carbon in all plant parts (kg C/ha)

lsu_plt_c is a state value: the monthly, yearly and average annual files give the value on the last day of the interval.

Layer count file

`carbon_layers.prt` is optional and is not listed in `file.cio`. When it is present in the run folder, it fixes N, the number of layer columns in every per-layer file. Use it to keep the same number of columns across runs, or to write only the top layers.

- Line 1: title, skipped.
- Line 2: header, skipped.
- Line 3: the number of layers, a whole number of 1 or more.

```
carbon_layers.prt: layers to expand in per-layer carbon output
n_layers
10
```

With `10`, every per-layer file has ten columns per variable, whatever the soils. A value below 1 is rejected with a message in `diagnostics.out`, and N then defaults to the largest layer count among the HRUs.

Legacy CSU carbon outputs

An older set of carbon files, kept for existing CSU workflows, runs beside the files above. SWAT+ Editor does not write the rows that switch them on; they are added to `print.prt` by hand. The code marks them for removal in revision 63.

`print.prt` row and flag	Files
`hru_cb` with `y` or `l` in the daily, monthly or yearly column	`hru_cbn_lyr.txt`, `hru_seq_lyr.txt`, `hru_n_p_pool_stat.txt`
as above, with `l` in any column	adds `hru_begsim_soil_prop.txt` and `hru_endsim_soil_prop.txt` (the start rows need `carbon = 2`)
as above, with `carbon = 2`	adds `hru_plc_stat.txt`, `hru_cflux_stat.txt`, `hru_cpool_stat.txt`
`hru_cb` with any flag other than `n`	`basin_carbon_all.txt` (header lines only when `hru_nb` has `y` in the average annual column)
`hru_cb_vars` with `y` or `l` in the daily, monthly or yearly column, and `carbon = 2`	`hru_carbvars.txt`, `hru_org_allo_vars.txt`, `hru_org_ratio_vars.txt`, `hru_org_trans_vars.txt`

These files have no interval suffix: one file collects the rows for every interval switched on. Each also gets a `.csv` copy when `csvout = y`. The legacy files contain the same quantities as the files above in a different layout. For example, legacy `hru_cbn_lyr.txt` and `hru_seq_lyr.txt` together contain the values that the new `hru_cbn_lyr_yr.txt` gives in one table.

The legacy files do not use the layer count file. The `cbn_diagnostics` setting from the old `carb_coefs.cbn` no longer exists; the `l` flag on the `hru_cb` row takes its place.

Example

Header and first row of `hru_carb_gl_yr.txt` from a two-year run of the `Ames_sub1` reference dataset with 62.0.1 (`carbon = 2`), shortened to the first eleven value columns:

```
      jday        mon        day         yr       unit             gis_id     name

       365         12         31       1975          1                  1 hru0001          0.702
```

Start of the first data row of `hru_cbn_lyr_yr.txt` from the same run, as values per column. Every soil in this dataset has 16 layers, so N = 16 and no column contains `-99`:

Column	Value
`unit`	1
`depth_lyr1` to `depth_lyr16`	10, 25, 50, 100, 150, 200, 300, 450, 600, 750, 900, 1050, 1200, 1500, 1800, 2000
`tot_300_sum`	44.06
`tot_lyr1`, `tot_lyr2`, `tot_lyr3`	2.28, 2.55, 4.24

The 12 HRU rows for 1975 appear 12 times each in this file (see Repeated rows).

Related files

- `print.prt`: output switches.
- `codes.bsn`: the `carbon` switch.
- `carbon.bsn` and `carbon_lyr.bsn`: carbon model inputs.
- Carbon model theory: pools and transformations.
- Naming convention: file suffixes and leading columns.

Sources

- `print.prt` rows: `basin_print_codes_read.f90`
- File opening and headers: `output_landscape_init.f90`, header types `output_carb_gl_header` and `output_hscf_header` in `output_landscape_module.f90`, variable lists and header and row helpers in `carbon_module.f90`
- Flat HRU files: `hru_carbon_output.f90` (units 4520 to 4527 and 4550 to 4557)
- Per-layer files: `soil_nutcarb_write.f90` and `soil_carbvar_write.f90` (units 4530 to 4605), called from `command.f90` and, for the end-of-run snapshot, `hru_output.f90`
- Landscape unit files: `lsu_carbon_output.f90` (units 4750 to 4773), landscape unit count from `lsu_read_elements.f90`
- Layer count file: `carbon_layers_read.f90`
- Legacy files: `carbon_legacy_module.f90` (units 8348 to 8387)

`print.prt` row	Flag variable
`hru_cb_gl`, `hru_cb_trf`, `hru_cb_lyr`, `hru_cb_cpool`, `hru_cb_npool`	`pco%cb_gl_hru`, `pco%cb_trf_hru`, `pco%cb_lyr_hru`, `pco%cb_cpool_hru`, `pco%cb_npool_hru`
`hru_cb_plt`, `hru_cb_flux`, `hru_cb_drv`, `hru_cb_dyn`, `hru_cb_snap`	`pco%cb_plt_hru`, `pco%cb_flux_hru`, `pco%cb_drv_hru`, `pco%cb_dyn_hru`, `pco%cb_snap_hru`
`lsu_cb_gl`, `lsu_cb_trf`, `lsu_cb_plt`	`pco%cb_gl_lsu`, `pco%cb_trf_lsu`, `pco%cb_plt_lsu`
`hru_cb`, `hru_cb_vars` (legacy)	`pco%cb_hru`, `pco%cb_vars_hru`

Output columns	Fortran source
`hru_carb_gl`	`hsc_d`, `hrc_d`, `hpc_d` (`carbon_soil_gain_losses`, `carbon_residue_gain_losses`, `carbon_plant_gain_losses`); `res_decay` is `rsd_decay_c`, `man_graze_c` is `man_graz_c`, the three `*_emit_c` columns are each type's `emit_c`
`hru_scf`	`hscf_d` (`carbon_soil_transformations`)
`tot_lyrK`, `seq_lyrK`, `tot_300_sum`, `seq_300_sum`	`soil1%tot(k)%c`, `soil1%seq(k)%c`, `soil1%tot_300_c`, `soil1%seq_tot_300_c`, all divided by 1000
`hru_cflux_stat`	`soil1%org_flx_lr(k)`; `_aa` uses `soil1%org_flx_cum_lr(k)` divided by `yrs_prt`
`hru_carb_drv`	`soil1%org_con_lr(k)`, `soil%ly(k)%tillagef*`, `soil%ly(k)%bmix`, `soil%phys(k)%tmp`, `soil1%emix(k)`
`hru_carb_dyn`	`soil1%org_allo_lr(k)`, `soil1%org_ratio_lr(k)`, `soil1%org_tran_lr(k)`
`lsu_plt_c`	sum of `pl_mass%tot_com%c * lsu_elem%ru_frac`